

03/04/2025

CODE-A



Aakash

Medical | IIT-JEE | Foundations

Corporate Office: AESL, 3rd Floor, Incuspaze Campus-2, Plot no. 13, Sector-18,
Udyog Vihar, Gurugram, Haryana-122015



AIM - 720

(Advanced INTENSIVE Mastery for 720)

MM : 720

PST-4

Time : 180 Mins.

Physics: System of Particles and Rotational Motion, Gravitation, Mechanical Properties of Solids, Mechanical Properties of Fluids

Chemistry: Organic Chemistry - Some Basic Principles and Techniques, Hydrocarbons, Thermodynamics

Botany: Photosynthesis in higher plants, Respiration in plants, Plant growth and development, Cell cycle and cell division

Zoology: Breathing and Exchange of gases, Body Fluids and Circulation, Excretory Products and their Elimination

Instructions:

- Duration of Test is 3 hrs.
- The Test consists of 180 questions. The maximum marks are 720.
- There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
- Each question carries +4 marks. For every wrong response, -1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
- Use blue/black ballpoint pen only to darken the appropriate circle.
- Mark should be dark and completely fill the circle.
- Dark only one circle for each entry.
- Dark the circle in the space provided only.
- Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

6 Aakashians scored a perfect 720/720 from Classroom Programs



720/720 | **Mridul M Anand**
3 Year Classroom



720/720 | **Ayush Naugraiya**
4 Year Classroom



720/720 | **Arghyadeep Dutta**
2 Year Classroom



720/720 | **Aryan Yadav**
1 Year Classroom

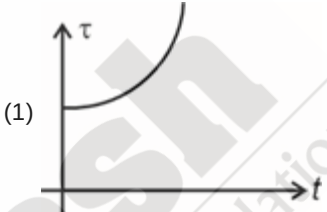
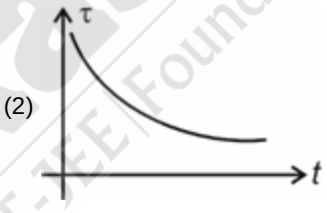
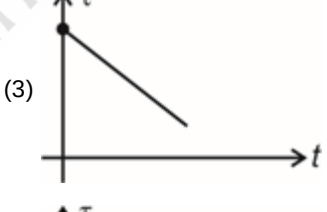
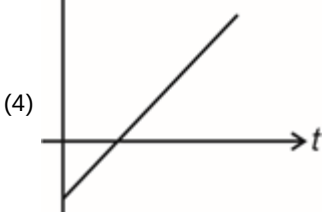


720/720 | **Palansha Agarwal**
2 Year Classroom

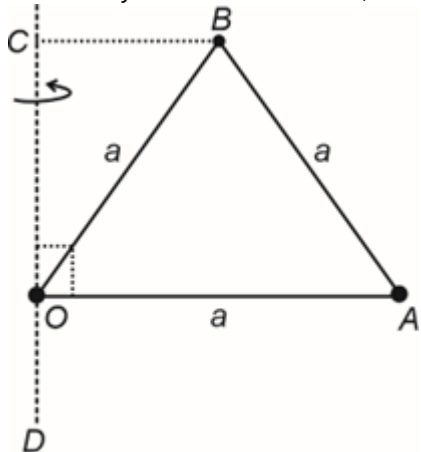


720/720 | **Iram Quazi**
1 Year Classroom

PHYSICS

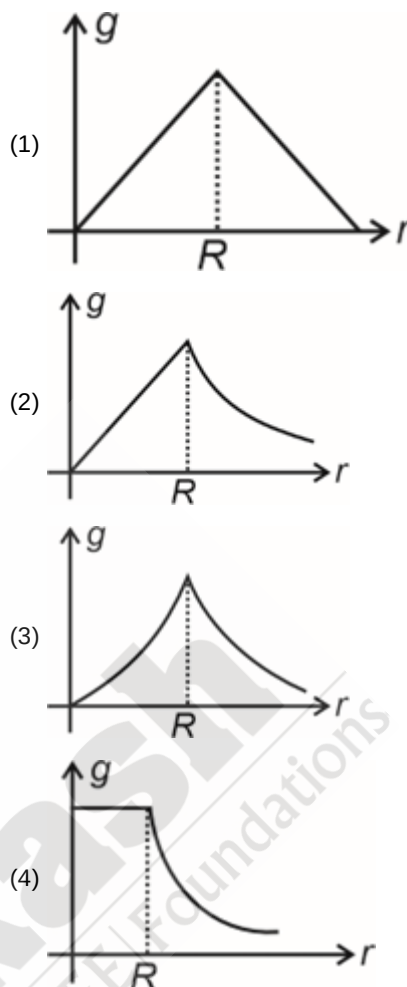
- A satellite of mass m moves from an orbit of radius r to $\frac{4}{3}r$, then change in total energy of satellite will be (Assume the mass of earth to be M)
 - $\frac{3GMm}{4r}$
 - $\frac{GMm}{2r}$
 - $\frac{GMm}{8r}$
 - $\frac{GMm}{4r}$
- If the ratio of acceleration due to gravity on surface of earth and mars is α and ratio of their radii is β , then ratio of respective escape speeds on their surface is
 - $\sqrt{2\alpha\beta}$
 - $\sqrt{\alpha\beta}$
 - $\sqrt{\frac{\alpha}{\beta}}$
 - $\sqrt{\frac{\alpha\beta}{2}}$
- When net external force on a system is zero, then
 - Velocity of centre of mass remains unchanged
 - Total linear momentum of system remains unchanged
 - Total linear momentum of system may change
 - Both (1) and (2)
- If two vector is given as $\vec{A} = \hat{i} - 2\hat{j} + 6\hat{k}$ and $\vec{B} = \hat{i} - 2\hat{j} + \hat{k}$, then the vector product $(\vec{A} \times \vec{B})$ will be equal to
 - $5\hat{i} + 10\hat{j}$
 - $5\hat{i} - 10\hat{j}$
 - $10\hat{i} + 5\hat{j}$
 - $10\hat{i} - 5\hat{j}$
- Consider the following statements out of which one is labelled as Assertion (A) and other as Reason (R).
Assertion (A): The centre of mass of body may lie where there is no mass.
Reason (R): Centre of mass of a system lie nearer to lighter part of the system.
 In the light of above statements, choose the **correct** option.
 - Both (A) and (R) are true and (R) is correct explanation of (A).
 - Both (A) and (R) are true but (R) is not correct explanation of (A).
 - Only (A) is true but (R) is false.
 - Both (A) and (R) are false.
- The angular speed (ω) of a body having moment of inertia I , changes according to expression, $\omega = 2(t - t^2)$. The variation of torque (τ) with time (t) is best represented by
 - 
 - 
 - 
 - 

7. Three-point masses each of mass m are placed at the vertices of an equilateral triangle of side a . The moment of inertia of system about axis COD , will be



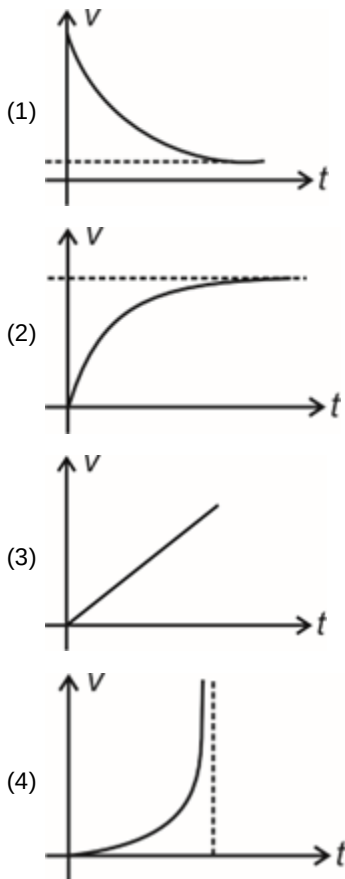
- (1) $2ma^2$
 (2) $\frac{2}{3}ma^2$
 (3) $\frac{5}{4}ma^2$
 (4) $\frac{7}{4}ma^2$
8. A wheel is rotating at the rate of 33 rev/min. If it comes to stop in 20 s, then magnitude of angular acceleration of wheel will be
- (1) $\pi \text{ rad/s}^2$
 (2) $11\pi \text{ rad/s}^2$
 (3) $\frac{11\pi}{200} \text{ rad/s}^2$
 (4) $\frac{\pi}{200} \text{ rad/s}^2$
9. Two particles of mass m and $9m$ are placed at a distance R . The gravitational potential on line joining the particles where the gravitational field equals zero will be
- (1) $\frac{-20Gm}{R}$
 (2) $\frac{-8Gm}{R}$
 (3) $\frac{-12Gm}{R}$
 (4) $\frac{-16Gm}{R}$
10. A body of mass 60 g experiences a gravitational force of 3 N, when placed at particular point of a hypothetical planet. The magnitude of gravitational field intensity at that point will be
- (1) 180 N/kg
 (2) 20 N/kg
 (3) 50 N/kg
 (4) 100 N/kg

11. Which graph among the given options shows the most appropriate variation of acceleration due to gravity (g) with distance from centre of earth (where R is radius of earth)

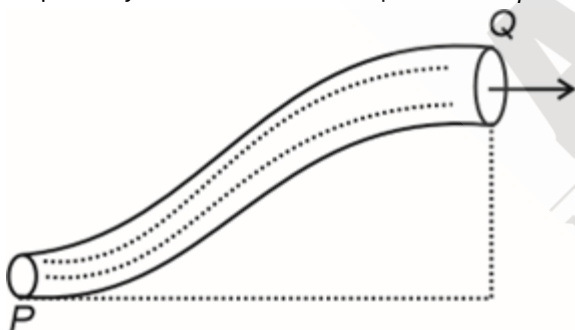


12. A liquid wets the solid surface if angle of contact is
- (1) Equal to 120°
 (2) Equal to 150°
 (3) Less than 90°
 (4) Equal to 100°

13. A spherical ball is dropped in a long column of a highly viscous liquid. The curve which shows the most appropriate variation of speed of ball (v) as a function of time (t) will be



14. An ideal fluid is flowing in a non-uniform cross sectional tube PQ (as shown in figure) from end P to end Q . If K_P and K_Q are kinetic energies per unit volume of fluid at P and Q respectively, the correct relationship between K_P and K_Q is



- (1) $K_P = K_Q$
 (2) $2K_P = K_Q$
 (3) $K_P > K_Q$
 (4) $K_P < K_Q$

15. The venturi-meter works on

- (1) Bernoulli's principle
 (2) Pascal's law
 (3) Huygen's principle
 (4) Stoke's law

16. Which of the following quantities remain constant in motion of a planet around sun in an elliptical orbit?

- (1) Speed
 (2) Angular speed
 (3) Kinetic energy
 (4) Angular momentum

17. Assuming the earth to be a sphere of uniform mass density, how much would a body weigh halfway down from the surface of earth if it weighs 250 N on surface?

- (1) 250 N
 (2) 125 N
 (3) 175 N
 (4) 145 N

18. The bulk modulus of water is $2 \times 10^9 \text{ N/m}^2$. The pressure required to increase the density of water by 0.1% is

- (1) $2 \times 10^3 \text{ N/m}^2$
 (2) $2 \times 10^6 \text{ N/m}^2$
 (3) $2 \times 10^5 \text{ N/m}^2$
 (4) $2 \times 10^7 \text{ N/m}^2$

19. A uniform steel rod of cross-sectional area A , uniform mass density ρ and length L is suspended so that it hangs vertically. The stress at mid point of rod is

- (1) $\frac{\rho g L}{4}$
 (2) $\frac{\rho g L}{2}$
 (3) $\frac{\rho g L}{A}$
 (4) $\frac{\rho g L}{2A}$

20. Two wires of same material have length and radius as (L, r) and $(2L, \frac{r}{2})$. If they are subjected to same load then, the ratio of their young's modulus will be

- (1) 1 : 2
 (2) 2 : 3
 (3) 4 : 1
 (4) 1 : 1

21. A rod elongates by l when a body of mass M is suspended from it. The elastic potential energy stored in the wire will be

(1) $\frac{Mgl}{2}$
 (2) Mgl
 (3) $2Mgl$
 (4) Zero

22. A wire of length L , area of cross-section A is hanging from fixed support. The length of wire changes to L_1 , when a load of certain mass M_0 is suspended from its free end. If young's modulus of material is Y , then the correct value of M_0 will be

(1) $\frac{AY}{g} \left(\frac{L_1}{L} + 1 \right)$
 (2) $\frac{AY}{g} \left(\frac{L_1}{L} - 1 \right)$
 (3) $\frac{Ag}{Y} \left(1 - \frac{L_1}{L} \right)$
 (4) $\frac{Ag}{Y} \left(\frac{L_1}{L} + 1 \right)$

23. An ice skater having moment of inertia I rotating with angular speed ω suddenly open arms, which reduces her angular velocity to $\frac{3\omega}{4}$. The change in moment of inertia of dancer is

(1) $\frac{I}{3}$
 (2) $\frac{4I}{3}$
 (3) $\frac{2I}{3}$
 (4) Zero

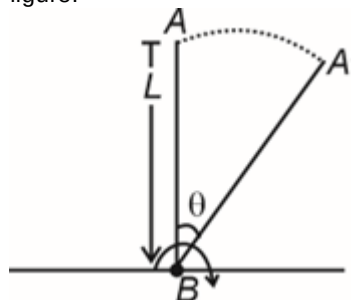
24. A force, $\vec{F} = 3\hat{j}$ N acts at origin. The torque due to this force about a point whose position vector is $2\hat{k}$ m, will be

(1) $6\hat{j}$ N m
 (2) $6\hat{i}$ N m
 (3) $-6\hat{j}$ N m
 (4) $-6\hat{k}$ N m

25. A constant torque of 100 N m turns a wheel having moment of inertia 300 kg m² about an axis passing through centre. Starting from rest, its angular velocity after 2 s will be

(1) $\frac{2}{3}$ rad/s
 (2) $\frac{3}{2}$ rad/s
 (3) 1 rad/s
 (4) 5 rad/s

26. A uniform rod of length L is free to rotate in a vertical plane about a fixed horizontal axis passing through B as shown in figure.



The rod begins rotating from rest. The angular velocity (ω) at angle θ is given by

(1) $\sqrt{\frac{6g}{L}} \sin \left(\frac{\theta}{2} \right)$
 (2) $\sqrt{\frac{6g}{L}} \cos \left(\frac{\theta}{2} \right)$
 (3) $\sqrt{\frac{6g}{L}} \sin \theta$
 (4) $\sqrt{\frac{6g}{L}} \cos \theta$

27. Two particles A and B are initially at rest, move towards each other under mutual force of attraction. At an instance when speed of A is v , then the speed of B if mass of particle A is m while that of particle B is $2m$, will be

(1) v
 (2) $2v$
 (3) $\frac{v}{2}$
 (4) $3v$

28. A satellite is orbiting just above the surface of earth with time period T . If d is density of the earth and G is universal constant of gravitation, then the correct expression for time period, T will be

(1) $\sqrt{\frac{3\pi}{Gd}}$
 (2) $\sqrt{\frac{4\pi}{3Gd}}$
 (3) $\sqrt{\frac{4Gd\pi}{3}}$
 (4) $\sqrt{\frac{AGd}{3\pi}}$

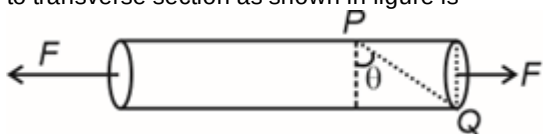
29. In a gravitational field, the gravitational potential is given by, $V = \frac{-K}{x^2}$ J/kg (where $x \neq 0$). The gravitational field intensity at point (2, 0, 3) m is

(1) $\frac{+K}{4} \hat{i}$
 (2) $\frac{+K}{2} \hat{i}$
 (3) $\frac{-K}{2} \hat{i}$
 (4) $\frac{-K}{4} \hat{i}$

30. The time period of a geostationary satellite is 24 h, at a height $6 R_E$ (R_E is radius of earth) from surface of earth. The time period of another satellite whose height is R_E from surface will be

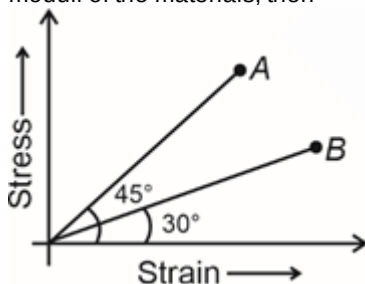
- (1) $\frac{48}{7} \sqrt{\frac{2}{7}} h$
- (2) $\frac{7}{48} \sqrt{\frac{7}{2}} h$
- (3) $\frac{48}{5} \sqrt{\frac{2}{5}} h$
- (4) $\frac{48}{5} \sqrt{\frac{5}{2}} h$

31. Two equal and opposite forces each of magnitude F is applied along a rod of transverse cross sectional area A . The tangential stress to a section PQ inclined at an angle θ to transverse section as shown in figure is



- (1) $\frac{F \sin \theta}{A}$
- (2) $\frac{F \cos \theta}{A}$
- (3) $\frac{F}{2A} \sin (2\theta)$
- (4) $\frac{F}{A} \cos^2 \theta$

32. The stress versus strain graph for wires of two materials A and B are as shown in figure. If Y_A and Y_B are the Young's moduli of the materials, then



- (1) $Y_A = \sqrt{3} Y_B$
- (2) $Y_B = \sqrt{3} Y_A$
- (3) $Y_A = Y_B$
- (4) $Y_A = 3 Y_B$

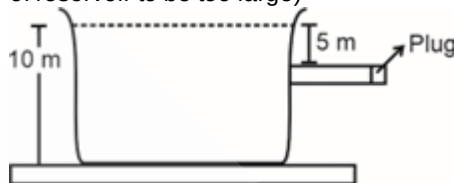
33. A steel wire of diameter 3 mm has a breaking strength of 4×10^5 N, then the breaking strength of similar steel wire of diameter 1.5 mm will be

- (1) 2.3×10^5 N
- (2) 1×10^5 N
- (3) 4.6×10^5 N
- (4) 3×10^5 N

34. A square glass plate of length 10 cm and thickness 0.4 cm, weighs 40 g in air. It is held vertically such that its lower edge rests on water surface. The difference between the apparent weight and actual weight of glass plate will be (Given surface tension of water = 0.073 N m^{-1})

- (1) 0.392 N
- (2) 0.407 N
- (3) 0.015 N
- (4) 0.075 N

35. As shown below, in a reservoir is 10 m deep. A horizontal pipe having cross-sectional area of 10^{-3} m^2 passes through reservoir 5 m below the water surface as shown in figure. A plug secures the pipe opening. (Assume the area of reservoir to be too large)



Based on above information, match the entries in column-I and column-II.

	Column-I		Column-II
(A)	The force (in N) required to hold plug in place.	(P)	0.01
(B)	The velocity of efflux (in m/s) as soon as the plug is removed	(Q)	50
(C)	The volume flow rate of water (in m^3/s) that flow out of pipe as soon as the plug is removed	(R)	10

- (1) A $\rightarrow$ Q, B $\rightarrow$ R, C $\rightarrow$ P
- (2) A $\rightarrow$ Q, B $\rightarrow$ P, C $\rightarrow$ R
- (3) A $\rightarrow$ R, B $\rightarrow$ Q, C $\rightarrow$ P
- (4) A $\rightarrow$ R, B $\rightarrow$ P, C $\rightarrow$ Q

36. Consider the following statements.

Statement (A): When two masses come closer, their gravitational potential energy decreases.

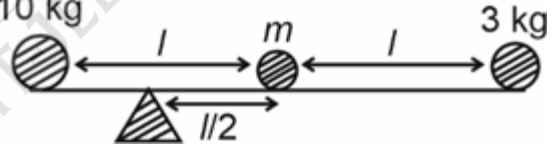
Statement (B): If linear momentum of an object is constant, then its angular momentum about any point remains constant.

Statement (C): Pressure is not a vector quantity.

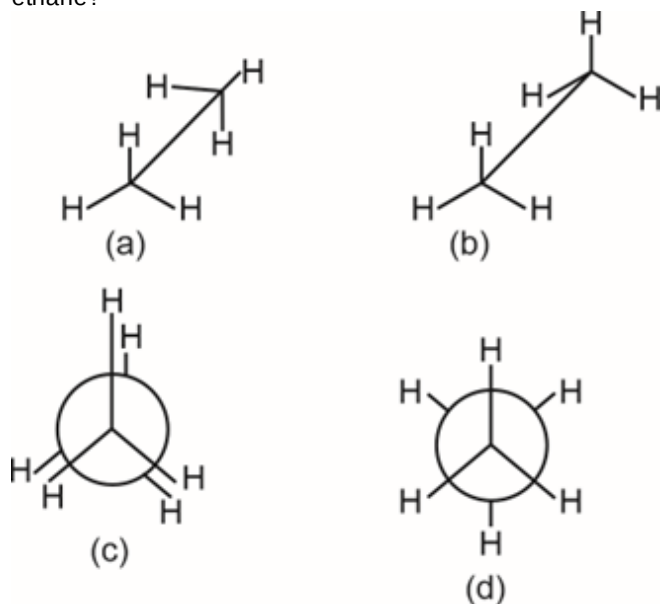
Statement (D): Centre of mass and centre of gravity may coincide if gravitational field is uniform.

Based on above information, pick the correct option.

- (1) Only statements (A) and (B) are correct.
- (2) Only statements (B) and (C) are correct.
- (3) Only statements (A) and (D) are correct.
- (4) Only statements (A), (C) and (D) are correct.

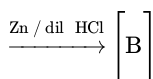
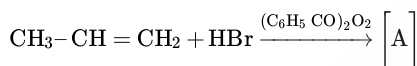
37. At what height from surface of earth, the acceleration due to gravity will be $\frac{1}{4}$ th of the value of g on the surface of earth.
- R
 - $2R$
 - $\frac{R}{4}$
 - $\frac{3R}{4}$
38. The correct dimensional formula for universal gravitational constant is
- $[M^{-1}L^3T^{-2}]$
 - $[ML^3T^{-2}]$
 - $[M^{-1}L^3T^2]$
 - $[M^{-1}L^{-3}T^{-2}]$
39. If two pieces of metal when immersed in a liquid have equal upthrust on them, then
- Both pieces must have equal weight
 - Both pieces must have equal densities
 - Both pieces must have equal volumes
 - Both (1) and (2)
40. A vessel contains a liquid which has a constant acceleration a_0 in horizontal direction. The free surface of water gets sloped with horizontal at an angle of (where g is acceleration due to gravity)
- $\tan^{-1}\left(\frac{a_0}{g}\right)$
 - $\tan^{-1}\left(\frac{g}{a_0}\right)$
 - $\cos^{-1}\left(\frac{a_0}{g}\right)$
 - $\sin^{-1}\left(\frac{g}{a_0}\right)$
41. A thin wire is bent in form of ring of radius r . The ring is placed horizontally on the surface of soap solution and then raised slowly. The upward force necessary to break the vertical film formed between ring and solution is (Assume the surface tension to be T)
- $4\pi rT$
 - $2\pi rT$
 - $3\pi rT$
 - $5\pi rT$
42. The excess pressure inside a spherical drop of water is five times that of another drop, then their respective surface area ratio will be
- 5 : 3
 - 25 : 9
 - 25 : 1
 - 25 : 137
43. A viscous liquid having coefficient of viscosity η , flows between two parallel plates along the x-axis. The difference between the velocity of two layers separated by distance dy is dv . If A is the area of each plate and F denotes the viscous force, then Newton's law of viscosity may be written as
- $\frac{F}{A} = -\eta \times \frac{dy}{dv}$
 - $\frac{F}{A} = -\eta \times \frac{dv}{dy}$
 - $F = \frac{-\eta}{A} \times \frac{dv}{dy}$
 - $F = \frac{-\eta}{A} \times \frac{dy}{dv}$
44. The maximum load that a wire can sustain is W . If the wire is cut to half its length, the maximum load it can sustain is
- W
 - $\frac{W}{2}$
 - $\frac{W}{4}$
 - $2W$
45. For equilibrium of the system as shown in figure, value of mass m should be
- 
- 9 kg
 - 3 kg
 - 1 kg
 - 5 kg

46. Which among the following is/are eclipsed conformation of ethane?



Choose the correct option.

- (1) (a) and (c) only
 (2) (b) and (c) only
 (3) (a) and (d) only
 (4) (b) and (d) only
47. Given below is the sequence of the reactions, major product [B] is



- (1) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{OH}$
 (2) $\text{CH}_3-\text{CH}_2-\text{CH}_3$
 (3) $\text{CH}_3-\underset{\text{OH}}{\text{CH}}-\text{CH}_3$
 (4) $\text{CH}_3-\text{CH}=\text{CH}_2$

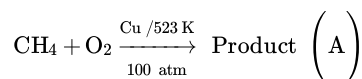
48. On electrolysis of sodium salt of acetic acid, the product obtained at anode is/are

- (a) CO_2
 (b) H_2
 (c) C_2H_6

Choose the correct option.

- (1) (a) and (c) only
 (2) (c) only
 (3) (a) and (b) only
 (4) (a), (b) and (c)

49. Consider the given reaction



Major product (A) is

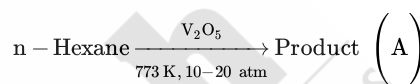
- (1) HCHO
 (2) CH_3COOH
 (3) HCOOH
 (4) CH_3OH

50. Consider the given compounds

- (a) Pentane
 (b) 2-Methylbutane
 (c) 2,2-Dimethylpropane
 The correct order of boiling point of the above compound is,

- (1) (a) > (c) > (b)
 (2) (b) > (a) > (c)
 (3) (a) > (b) > (c)
 (4) (c) > (b) > (a)

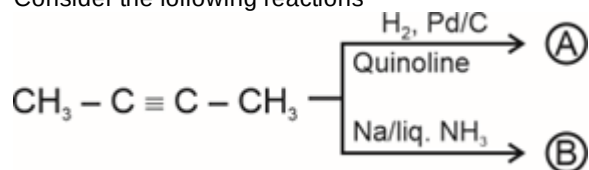
51. Consider the following reaction,



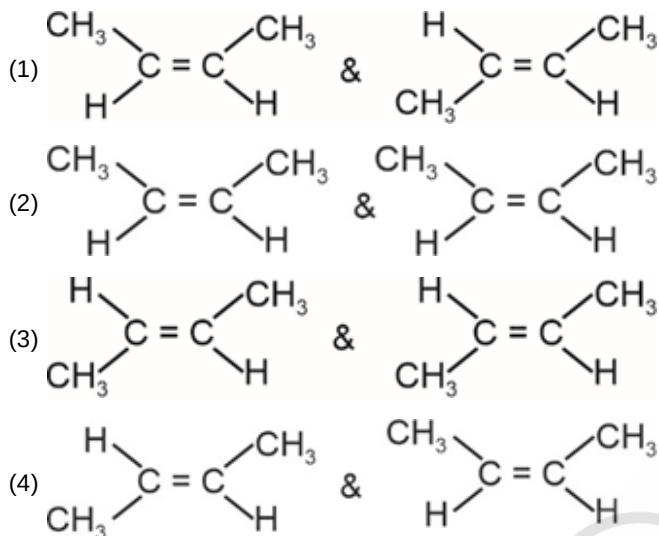
Major product (A) is

- (1)
- (2)
- (3)
- (4)

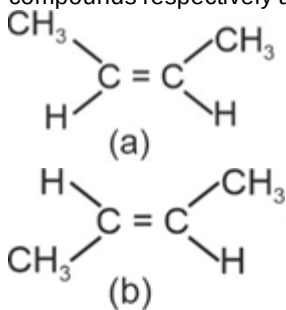
52. Consider the following reactions



Major product (A) and (B) respectively are,

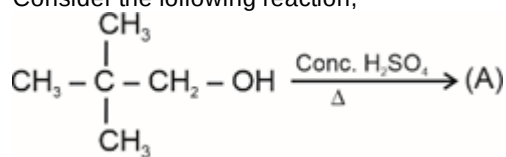


53. The correct order of boiling point and melting point of given compounds respectively are

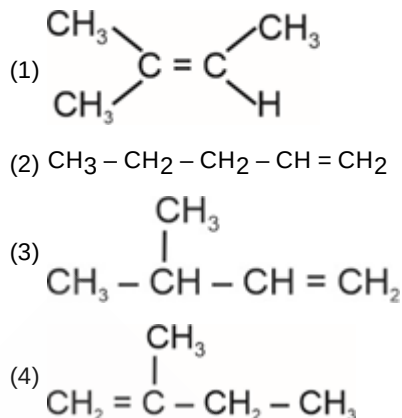


- (1) (a) > (b) ; (a) > (b)
 (2) (b) > (a) ; (a) > (b)
 (3) (a) > (b) ; (b) > (a)
 (4) (b) > (a) ; (b) > (a)

54. Consider the following reaction,



Major product (A) is



55. Match list-I with list-II and choose the correct option.

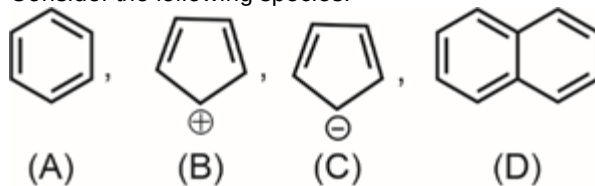
**List-I
(Reactions)**

**List-II
(Relation between ΔH and ΔU)**

- a. $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$ (i) $\Delta H = \Delta U - \frac{1}{2}RT$
 b. $\text{PCl}_5(\text{g}) \rightarrow \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ (ii) $\Delta H = \Delta U + RT$
 c. $\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$ (iii) $\Delta H = \Delta U - 2RT$
 d. $\text{CO}(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$ (iv) $\Delta H = \Delta U$

- (1) a(iii), b(iv), c(i), d(ii)
 (2) a(ii), b(iii), c(iv), d(i)
 (3) a(ii), b(iv), c(i), d(iii)
 (4) a(iii), b(ii), c(iv), d(i)

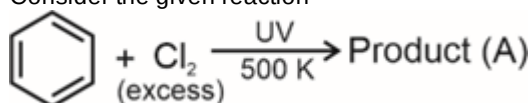
56. Consider the following species.



The aromatic species are

- (1) (A), (B) and (C) only
 (2) (A) and (B) only
 (3) (C) and (D) only
 (4) (A), (C) and (D) only

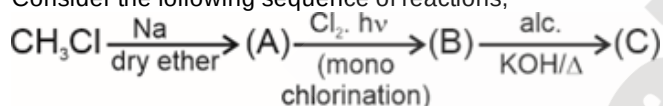
57. Consider the given reaction



Major product (A) is

- (1)
- (2)
- (3)
- (4)

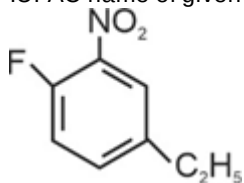
58. Consider the following sequence of reactions,



Major product (C) is

- (1) CH_3CHO
 (2) $\text{CH}_2 = \text{CH}_2$
 (3) $\text{CH}_3 - \text{OH}$
 (4) HCOOH

59. IUPAC name of given compound is



- (1) 2-Ethyl-5-fluoro-6-nitrobenzene
 (2) 3-Ethyl-6-fluoro-1-nitrobenzene
 (3) 4-Ethyl-1-fluoro-2-nitrobenzene
 (4) 1-Ethyl-4-fluoro-3-nitrobenzene

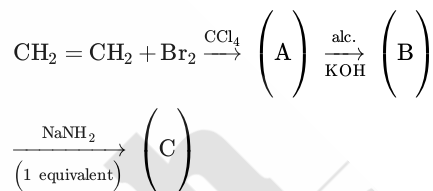
60. The correct order of stability of given radicals is

- (1) $\cdot\text{CH}_3 > \cdot\text{CH}_2\text{CH}_3 > \cdot\text{CH}(\text{CH}_3)_2 > \cdot\text{C}(\text{CH}_3)_3$
 (2) $\cdot\text{CH}_2\text{CH}_3 > \cdot\text{CH}_3 > \cdot\text{CH}(\text{CH}_3)_2 > \cdot\text{C}(\text{CH}_3)_3$
 (3) $\cdot\text{C}(\text{CH}_3)_3 > \cdot\text{CH}(\text{CH}_3)_2 > \cdot\text{CH}_2\text{CH}_3 > \cdot\text{CH}_3$
 (4) $\cdot\text{C}(\text{CH}_3)_3 > \cdot\text{CH}_2\text{CH}_3 > \cdot\text{CH}(\text{CH}_3)_2 > \cdot\text{CH}_3$

61. In Carius method of estimation of halogen, 0.10 g of an organic compound gave 0.08 g of AgBr . The percentage of bromine in organic compound is

- (1) 23.5%
 (2) 69.3%
 (3) 42.5%
 (4) 34.0%

62. Consider the following sequence of reactions,



Major product (C) is

- (1) $\text{H}_2\text{C} = \text{CH}_2$
 (2) $\text{CH}_2 = \text{CHBr}$
 (3) $\text{HC} \equiv \text{CH}$
 (4) $\text{CH}_3\text{CH}_2\text{NH}_2$

63. The correct order of decreasing priority of given functional groups in IUPAC nomenclature is

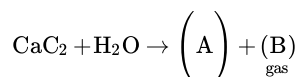
- (1) $-\text{SO}_3\text{H} > -\text{COOR} > -\text{COOH} > -\text{CHO}$
 (2) $-\text{COOR} > -\text{SO}_3\text{H} > -\text{CHO} > -\text{COOH}$
 (3) $-\text{COOH} > -\text{SO}_3\text{H} > -\text{COOR} > -\text{CHO}$
 (4) $-\text{COOR} > -\text{SO}_3\text{H} > -\text{COOH} > -\text{CHO}$

64. The sodium fusion extract is boiled with iron (II) sulphate and then acidified with concentrated sulphuric acid. The formation of Prussian blue colour confirms the presence of nitrogen.

Prussian blue colour is due to the formation of

- (1) $\text{Na}_2[\text{Fe}(\text{CN})_5(\text{H}_2\text{O})]$
 (2) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3 \cdot x\text{H}_2\text{O}$
 (3) $\text{Fe}(\text{SCN})_3$
 (4) $\text{Na}_3[\text{Fe}(\text{CN})_5(\text{H}_2\text{O})]$

65. Consider the given reaction,



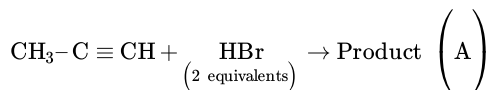
The product (B) is

- (1) H_2
- (2) C_2H_2
- (3) C_2H_6
- (4) CH_4

66. On complete combustion, 0.126 g of an organic compound gave 0.099 g of carbon dioxide and 0.0507 g of water. The percentage composition of carbon in the organic compound is

- (1) 63.27%
- (2) 21.42%
- (3) 42.84%
- (4) 15.86%

67. Consider the following reaction



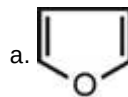
Major product (A) is

- (1)
- (2)
- (3)
- (4)

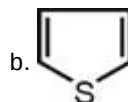
68. Match List-I with List-II and choose the correct option.

List-I
(Compounds)

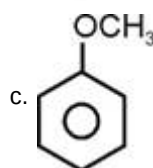
List-II
(Common names)



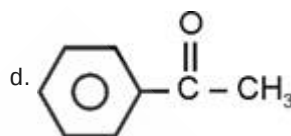
(i) Acetophenone



(ii) Anisole



(iii) Thiophene



(iv) Furan

(1) a(i), b(iii), c(ii), d(iv)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(iv), b(ii), c(i), d(iii)

(4) a(iv), b(iii), c(ii), d(i)

69. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): For an adiabatic expansion process, work done by the system is equal to change in internal energy

Reason (R): There is no heat exchange between system and surrounding in an adiabatic process

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

70. When 10 moles of an ideal gas is compressed reversibly from 100 L to 10 L at constant temperature. The entropy change during the process is

(1) +46.06 R

(2) -R

(3) -23.03 R

(4) -46.06R

71. Given below are two statements:

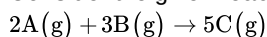
Statement I: Both heat and work are state function.

Statement II: Pressure and heat capacity are intensive properties

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Both Statement I and Statement II are correct
- (2) Both Statement I and Statement II are incorrect
- (3) Statement I is correct but Statement II is incorrect
- (4) Statement I is incorrect but Statement II is correct

72. Consider the given reaction



The change in internal energy is -10 kJ and change in entropy is -30 J K^{-1} , the change in Gibbs free energy at 300 K is

- (1) $+19000 \text{ J}$
- (2) -1000 J
- (3) -19000 J
- (4) $+1000 \text{ J}$

73. The correct option for free expansion of an ideal gas under isothermal condition is

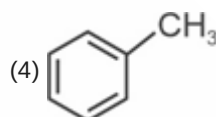
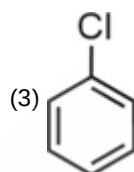
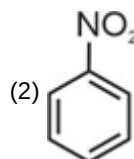
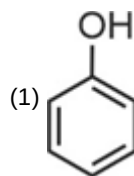
- (1) $q = 0, w < 0, \Delta U > 0$
- (2) $q < 0, w < 0, \Delta U > 0$
- (3) $q = 0, w = 0, \Delta U = 0$
- (4) $q > 0, w > 0, \Delta U > 0$

74. Match thermodynamic condition in list-1 with spontaneity condition in list-2 and choose the correct answer

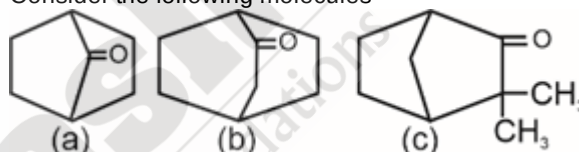
List-1	List-2
(a) $\Delta H > 0, \Delta S > 0$	(i) Spontaneous at all temperatures
(b) $\Delta H < 0, \Delta S < 0$	(ii) Non spontaneous at all temperatures
(c) $\Delta H > 0, \Delta S < 0$	(iii) Spontaneous, if $T > \frac{\Delta H}{\Delta S}$
(d) $\Delta H < 0, \Delta S > 0$	(iv) Spontaneous, if $T < \frac{\Delta H}{\Delta S}$

- (1) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)
- (2) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
- (3) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
- (4) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

75. Which among the following is most reactive towards electrophilic aromatic substitution reaction?



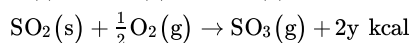
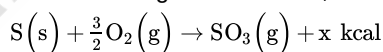
76. Consider the following molecules



The molecule(s) which will show tautomerism is/are

- (1) (a) and (b) only
- (2) (b) only
- (3) (a) and (c) only
- (4) (a), (b) and (c)

77. Consider the given reactions,



The heat of formation of SO_2 is

- (1) $x + y$
- (2) $(2y - x)$
- (3) $x + 2y$
- (4) $-(x + 2y)$

78. Given below are two statements:

Statement I: When a system is at equilibrium, the entropy is maximum.

Statement II: When a system is at equilibrium, ΔS_{total} is zero

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Both Statement I and Statement II are incorrect
- (2) Both Statement I and Statement II are correct
- (3) Statement I is correct but Statement II is incorrect
- (4) Statement I is incorrect but Statement II is correct

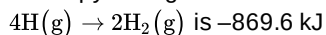
79. Consider the following statements,

- (a) On conversion of diamond to graphite, entropy increases
- (b) If ΔG is positive, the reaction is non spontaneous
- (c) $\frac{q_{\text{rev}}}{T}$ is independent of path

The correct statement(s) is/are

- (1) (a) and (b) only
- (2) (a) and (c) only
- (3) (c) only
- (4) (a), (b) and (c)

80. Enthalpy change for the reaction,



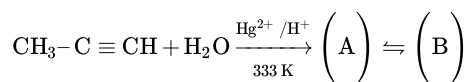
The dissociation energy of H-H bond is

- (1) 157.6 kJ mol⁻¹
- (2) 434.8 kJ mol⁻¹
- (3) 217.4 kJ mol⁻¹
- (4) 326.5 kJ mol⁻¹

81. In a process, 600 J of heat is absorbed by the system and 250 J of work is done by the system, change in internal energy during the process is

- (1) 100 J
- (2) 850 J
- (3) 350 J
- (4) 250 J

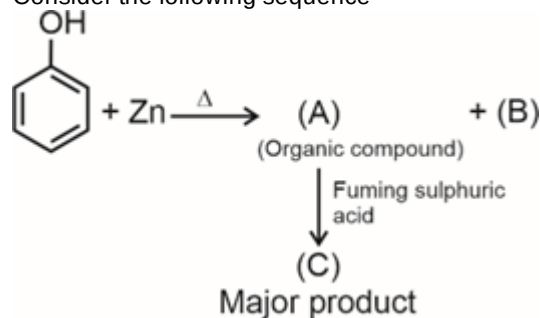
82. Consider the following reaction,



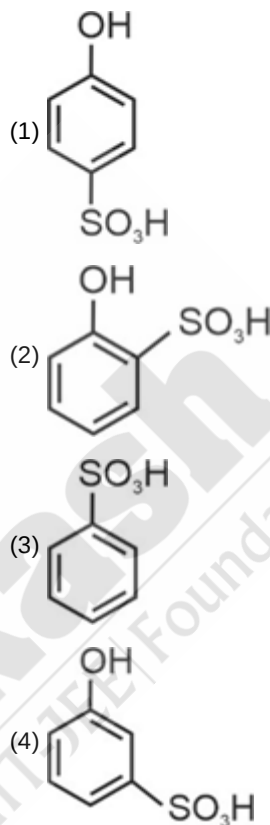
Major product (B) is

- (1) CH₃CH₂CHO
- (2) CH₃CH₂CH₂OH
- (3) CH₃COCH₃
- (4) CH₃CH₂COOH

83. Consider the following sequence



Major product (C) is



84. In Dumas method for estimation of nitrogen, 0.16g of an organic compound gave 46 mL of N₂ measured over H₂O at 15°C and at a pressure of 755 mm Hg. Percentage of nitrogen in given compound is (approx.)
(Given aqueous tension of H₂O = 15 mm of Hg)

- (1) 59%
- (2) 33%
- (3) 3.4%
- (4) 14%

85. Given below are two statements:

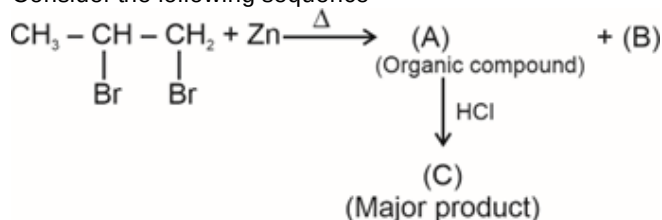
Statement I: Kjeldahl's method is not applicable to compounds containing nitrogen in nitro compounds.

Statement II: Percentage composition of halogens are estimated using carius method

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Both Statement I and Statement II are incorrect
- (2) Both Statement I and Statement II are correct
- (3) Statement I is correct but Statement II is incorrect
- (4) Statement I is incorrect but Statement II is correct

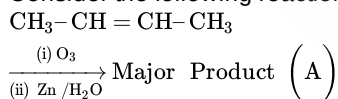
86. Consider the following sequence



Major product (C) is

- (1) $\text{CH}_3 - \underset{\text{Cl}}{\text{CH}} - \text{CH}_3$
- (2) $\text{Cl} - \text{CH}_2 - \text{CH} = \text{CH}_2$
- (3) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2\text{Cl}$
- (4) $\text{CH}_3 - \underset{\text{Cl}}{\text{CH}} - \underset{\text{Br}}{\text{CH}_2}$

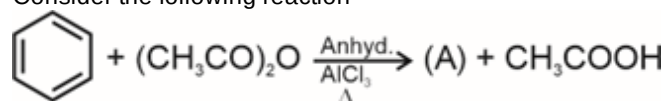
87. Consider the following reaction,



Major product (A) is

- (1) HCHO
- (2) CH₃CHO
- (3) CH₃CH₂CH₂CHO
- (4) CH₃COOH

88. Consider the following reaction



Major product (A) is

- (1)
- (2)
- (3)
- (4)

89. Given below are two statements:

Statement I: Rate of reaction of alkanes with halogens follows the order, F₂ > Cl₂ > Br₂ > I₂.

Statement II: Halogenation of alkanes proceeds via free radical mechanism.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Both Statement I and Statement II are incorrect
- (2) Both Statement I and Statement II are correct
- (3) Statement I is correct but Statement II is incorrect
- (4) Statement I is incorrect but Statement II is correct

90. Given below are two statements:

Statement I: Column chromatography is based on the principle of differential adsorption.

Statement II: Thin layer chromatography is a type of partition chromatography.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Both Statement I and Statement II are incorrect
- (2) Both Statement I and Statement II are correct
- (3) Statement I is incorrect but Statement II is correct
- (4) Statement I is correct but Statement II is incorrect

BOTANY

91. The process which makes major difference between tomato and maize plants is

- (1) Glycolysis
- (2) Calvin cycle
- (3) Photorespiration
- (4) Respiration

92. In mitochondrial matrix, the enzyme that catalyses the conversion of end product of glycolysis into acetyl CoA requires

- (1) Magnesium ions
- (2) Cadmium ions
- (3) Ferric ions
- (4) Cobalt ions

93. In T.W. Engelmann experiment, *Cladophora* a green alga was placed in a suspension of aerobic bacteria. The purpose of using aerobic bacteria was to

- (1) Measure the amount of light absorbed by *Cladophora* at different wavelengths.
- (2) Detect the sites of oxygen evolution by the green algae.
- (3) Measure the amount of glucose stored as starch
- (4) Identify the part of the algae that could release nitrogen.

94. Which combination of hormones can be applied to artificially induce flowering in the pineapple plant?

- (1) GA₃ and ABA
- (2) Auxin and C₂H₄
- (3) Kinetin and ABA
- (4) Ethephon and GA₃

95. The cell organelle that is responsible for aster formation in animal cell duplicates during

- (a) G₀ phase
- (b) G₁ phase
- (c) S phase
- (d) Interkinesis
- (e) G₂ Phase

The **correct** ones are

- (1) c and d only
- (2) a and b only
- (3) c and e only
- (4) b and d only

96. Which of the following statements is **not** true?

- (1) Absorption maxima for chlorophyll *b* and carotenoids is in the range of 400-500 nm.
- (2) Rate of O₂ release during photosynthesis is maximum when the action spectrum of photosynthesis is lowest.
- (3) Action spectrum of photosynthesis corresponding closely to the absorption spectra of chlorophyll *a* shows that it is the chief pigment.
- (4) Accessory pigments help to make photosynthesis more efficient by absorbing different wavelength of light

97. During the conversion of two molecules of glyceraldehyde-3-phosphate into 1, 3-bisphosphoglyceric acid. Which of the following is formed?

- (1) H₂O
- (2) ATP
- (3) NADH + H⁺
- (4) ADP

98. In most higher plants, the growing apical bud inhibits the growth of the lateral (axillary) buds. This phenomenon can be removed/inhibited by

- (1) Decapitation of the apical buds.
- (2) Treatment of apical bud with sterile filtrate of *Gibberella fujikuroi*.
- (3) Application of minute quantity of ABA to axillary buds.
- (4) Removal of the plant hormone like cytokinins from the apical buds.

99. Which of these statements is **incorrect**?

- (1) In anaerobic organisms, glycolysis is the only process in respiration.
- (2) Anaerobic respiration occurs in cytoplasm and mitochondria.
- (3) Oxidative phosphorylation takes place in inner mitochondrial membrane
- (4) Glycolysis operates as long as it is supplied with NAD⁺ that can pick up hydrogen atoms.

100. Some cells of adult animals like heart cells do not divide further, as they exit a phase of cell cycle to enter an inactive stage. That particular phase is

- (1) M Phase
- (2) S phase
- (3) G₁ phase
- (4) G₂ phase

101. Identify the **correct** statement regarding the S phase of interphase

- (1) Centrosome begins to move towards opposite poles of the cell
- (2) No cell organelle gets duplicated
- (3) DNA synthesis or replication takes place.
- (4) Nuclear division and separation of daughter chromosomes takes place

102. Arrange the events in a sequential manner that occur during Calvin cycle and mark the **correct** option.

- (a) Utilisation of ATP for regeneration of RuBP.
- (b) CO₂ combines with ribulose-1, 5-bisphosphate
- (c) Reduction of 3-phosphoglyceric acid

- (1) b → a → c
- (2) b → c → a
- (3) a → c → b
- (4) c → b → a

103. Which statement is **correct** for Krebs cycle?

- (1) It starts with the condensation of acetyl group with three carbon molecule and water to yield citric acid.
After isomerisation of citrate into isocitrate and then
- (2) oxalosuccinic acid two successive steps of decarboxylation occur.
- (3) There is only one point in the cycle where NAD⁺ is reduced to NADH + H⁺
- (4) The continued oxidation of acetyl CoA via the TCA cycle requires the continued replenishment of pyruvic acid

104. Identify the **incorrectly** matched pair.

(1)	Auxin	–	Stimulates root formation on stem cuttings
(2)	Gibberellin	–	Elongate and improves apple shape
(3)	Cytokinin	–	Induce dormancy of buds, seeds and storage organs
(4)	Ethylene	–	Promotes development of lateral roots and growth of root hairs

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

105. Match the Column-I with Column-II.

	Column-I		Column-II
(a)	Leptotene	(i)	Also known as dictyotene stage
(b)	Zygotene	(ii)	Condensation and coiling of chromatin fibres begins
(c)	Diakinesis	(iii)	Two homologous chromosomes form a pair
(d)	Diplotene	(iv)	Terminalization of chiasmata

Select the **correct** option from the following.

- (1) (a) (ii); (b) (i); (c) (iii); (d) (iv)
- (2) (a) (ii); (b) (iii); (c) (iv); (d) (i)
- (3) (a) (iii); (b) (ii); (c) (iv); (d) (i)
- (4) (a) (i); (b) (ii); (c) (iii); (d) (iv)

106. Prophase of mitosis can be characterised by

- (a) Untangling of chromosomes
- (b) Condensation of chromosomal material
- (c) Congression
- (d) Spireme stage
- (e) Formation of interzonal fibres

Select the **correct** ones

- (1) a, b, c and e
- (2) a, b and d
- (3) b, c and e
- (4) c, d and e

107. The complete disintegration of the nuclear envelope marks the end of

- (1) Metaphase
- (2) Anaphase
- (3) Prophase
- (4) Telophase

108. Select the **odd** one w.r.t. plant growth inhibitor.

- (1) Dormin
- (2) ABA
- (3) Zeatin
- (4) Abscission II

109. Which of the following is an adenine derivative?

- (1) Dormin
- (2) ABA
- (3) NAA
- (4) Kinetin

110. Along with auxins, extract from which components can be used to proliferate a mass of undifferentiated cells in a nutrient medium?

- (a) Meristematic tissues
- (b) Sterile filtrate of *Gibberella fujikuroi*
- (c) Yeast
- (d) Coconut milk
- (e) Autoclaved herring sperm DNA

- (1) Only a, b and e
- (2) All except b
- (3) Only b and e
- (4) Only c and d

111. Read the following Assertion (A) and Reason (R) statements and select the **correct** option.

Assertion (A) – During oxidation within a cell, all the energy contained in respiratory substrates is not immediately released free into the cell.

Reason (R) – Energy is released in a series of slow step-wise reactions controlled by enzymes and is trapped in form of ATP.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) Both (A) and (R) are false.
- (4) (A) is true but (R) is false

112. Both auxin and cytokinin promote

- (1) Cell division
- (2) Shedding of leaves
- (3) Seed dormancy
- (4) Senescence

113. Which plant growth regulator inhibits longitudinal growth but stimulates transverse growth of seedlings?

- (1) ABA
- (2) Gibberellins
- (3) Ethylene
- (4) Cytokinin

114. The first natural cytokinin was isolated from

- (1) Degraded product of autoclaved herring sperm DNA
- (2) Unripe maize grains or kernels
- (3) Human urine
- (4) Tips of coleoptiles of oat seedlings

115. It has been observed that the carpel producing parthenocarpic fruits show higher production of

- (1) Cytokinin
- (2) Auxins
- (3) Gibberellins
- (4) Ethylene

116. Cells at the shoot and root tip shows all the given features, **except**

- (1) Dense protoplasm
- (2) Large nucleus
- (3) Protoplasmic modification
- (4) Primary cell wall

117. With reference to the factors affecting the rate of photosynthesis, which of the following statements is **not** correct?

- (1) There is a linear relationship between incident light and CO_2 fixation at low light intensities.
- (2) Water is a major limiting factor influencing the rate of photosynthesis
- (3) Increasing atmospheric CO_2 concentration upto 0.05% can enhance CO_2 fixation rate.
- (4) Optimum temperature in C_3 plants is $20-25^\circ\text{C}$ and for C_4 plants is $30-45^\circ\text{C}$

118. Read the following statements and select the **correct** option.

Statement (A): Within the leaves, the mesophyll cells have a large number of chloroplasts that are responsible for CO_2 fixation.

Statement (B): Within chloroplasts, stroma is the site for light reaction, while the chemosynthetic pathway occurs in the membranes.

- (1) Both (A) and (B) are correct
- (2) Both (A) and (B) are incorrect
- (3) Only statement (B) is correct
- (4) Only statement (A) is correct

119. Select the **correct** statement w.r.t. chemiosmotic hypothesis.

- (1) As protons move through the photosystems, electrons are transported across the membrane.
- (2) The enzyme that reduces NADP^+ into NADPH, removes proton from the stroma.
- (3) Within the chloroplast, protons in the stroma always increase in number
 CF_1 particle of the ATP synthase is a transmembrane
- (4) channel that allows movement of protons across thylakoid membrane

120. The most dramatic period of the cell cycle involves

- (1) A major reorganisation of virtually all components of the cell
- (2) Restoration of chromosome number during sexual reproduction
- (3) Movement of homologous chromosome to the opposite poles of cell
- (4) Formation of dyad of cells and cell organelles

121. The complete disintegration of the nuclear envelope marks the start of a phase of the cell cycle that does not include

- (1) Attachment of spindle fibres to kinetochores
- (2) Completion of condensation of chromosomal material to form a compact mitotic chromosome
- (3) The alignment of chromosomes at the equator of the cell
- (4) Movement of centrosomes to opposite poles

122. Nuclear envelope develops around the chromosome clusters at each pole forming two daughter nuclei during

- (1) Anaphase
- (2) Metaphase
- (3) Prophase
- (4) Telophase

123. During cytokinesis, a furrow in the plasma membrane develops and divides the cell cytoplasm into two in

- (1) *Chlamydomonas*
- (2) White blood cells
- (3) Mature RBCs
- (4) *Cladophora*

124. In chloroplast, stroma lamellae lacks

- (1) PS I and NADP reductase enzyme
- (2) NADP reductase and P_680
- (3) Water splitting complex and PS I
- (4) Electron transport system and oxygen evolving complex

125. C_3 plants are similar to C_4 plants as both

- Have components that are necessary for carboxylation,
- (1) reduction and regeneration reaction for synthesis of starch
- (2) Have ability to adapt to dry tropical regions and higher temperatures
- (3) Have primary CO_2 acceptor as a 3-carbon molecule phosphoenolpyruvate present in mesophyll cells
- (4) Shows photosynthetic saturation at CO_2 concentration above $360 \mu\text{L}^{-1}$ in low light intensity.

126. Identify the **incorrect** statement w.r.t. C_4 pathway.

- (1) It requires the presence of mesophyll cells and bundle sheath cells.
- (2) The initial or the first product of this pathway is a four-carbon compound.
- (3) The pyruvic acid formed due to decarboxylation reaction is the primary CO_2 acceptor that forms C_4 acid.
- (4) PEP carboxylase is the enzyme that converts atmospheric CO_2 into oxaloacetic acid.

- 127.** In both lactic acid and alcohol fermentation,
- Less than seven percent of the energy in glucose is released
 - Net gain is 2ATP
 - Decarboxylation occurs, leading to CO₂ release
 - The reducing agent NADH + H⁺ is reoxidised to NAD⁺
 - Pyruvic acid is the common substrate for lactic acid and alcohol fermentation
- In the light of the above statement select the **correct** ones.

- a, b and d only
- b, c and e only
- a, b, d and e only
- c, d and e only

- 128.** Substrate level phosphorylation is

- Addition of phosphate group to ATP
- Formation of ATP by energy released from electrons removed during substrate oxidation
- Oxidation of phosphate group in ATP
- Formation of ATP by transfer of phosphate group from a substrate to ADP

- 129.** Select the **incorrectly** matched pair.

(1)	Complex I	—	NADH dehydrogenase
(2)	Complex II	—	Converts fumarate into succinate
(3)	Complex III	—	Ubiquinol is oxidised by transferring its electron to Cyt C
(4)	Complex IV	—	Cytochrome c oxidase

- (1)
- (2)
- (3)
- (4)

- 130.** Read the following statements and select the **correct** option.

Statement A : Gibberellins increase the length of axis.

Statement B : Gibberellic acids used to speed up the malting process in brewing industry.

- Both statements A and B are correct
- Both statements A and B are incorrect
- Only statement A is correct
- Only statement B is correct

- 131.** Read the following Assertion (A) and Reason (R) statements and select the **correct** option.

Assertion (A) : Growth is measured by a variety of parameters such as fresh weight, dry weight, length, area, volume and cell number but not by measuring the increase in protoplasm.

Reason (R) : Growth at cellular level is principally a consequence of increase in the amount of protoplasm and protoplasm is difficult to measure directly.

- Both (A) and (R) are true but (R) is not the correct explanation of (A)
- Both (A) and (R) are true and (R) is the correct explanation of (A)
- (A) is true but (R) is false
- Both (A) and (R) are false

- 132.** Select the **correct** sequential order of cell cycle.

- G₁, G₂, S, M
- M, S, G₂, G₁
- G₁, S, G₂, M
- S, G₁, G₂, M

- 133.** Mark **true** (T) or **false** (F) for the given statements and select the **correct** option.

- Identity of chromosome is lost as discrete elements during telophase.
- Animals never show equational division in haploid cells
- Second stage of prophase I is long-lived as compared to third stage
- Homologous chromosomes remain linked at the sites of crossing over by the end of pachytene

	(a)	(b)	(c)	(d)
(1)	T	F	F	T
(2)	T	T	F	T
(3)	F	T	T	F
(4)	T	F	T	F

- (1)
- (2)
- (3)
- (4)

134. Select the **incorrect** match from the following.

(1)	Pachytene	–	Exchange of genetic material between two homologous chromosomes
(2)	Recombinase	–	Activity is seen in Zygotene
(3)	Diplojene	–	Dissolution of synaptonemal complex
(4)	Diakinesis	–	Represent transition to metaphase

(1) (1)

(2) (2)

(3) (3)

(4) (4)

135. Select the **incorrect** statement w.r.t. meiosis.

- (1) In both diakinesis and prophase II, nucleolus and nuclear envelope disintegrate by the end of the stage
- (2) The bivalent chromosomes aligns on the equatorial plate during metaphase II
- (3) It increases the genetic variability in the population of organisms from one generation to the next
- (4) The homologous chromosomes separate, while sister chromatids remain associated at their centromeres during anaphase I

ZOOLOGY

136. In an adult human, under normal conditions, the value of pO_2 present in deoxygenated blood of pulmonary artery is equal to the value of pCO_2 in

- (1) Atmospheric air
- (2) Tissues
- (3) Alveoli
- (4) Deoxygenated blood flowing through systemic vein

137. Select the **correct** match w.r.t compatible blood groups of donor and recipient for blood transformation

	Donor	Recipient
(1)	O^-	A^+
(2)	AB^+	B^-
(3)	A^+	AB^-
(4)	B^+	O^-

(1) (1)

(2) (2)

(3) (3)

(4) (4)

138. The chronic disorder which is characterised by decreased respiratory surface and whose one of the major causes is cigarette smoking is

- (1) Asthma
- (2) Emphysema
- (3) Occupational respiratory disorder
- (4) Allergy

139. During swallowing of food, A is covered by epiglottis to prevent entry of food into B. Select the option which correctly represents A and B.

	A	B
(1)	Gullet	Sound box
(2)	Glottis	Food pipe
(3)	Pharynx	Food pipe
(4)	Glottis	Larynx

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

140. Which of the following options represents the correct set of factors which are favourable for the formation of oxyhaemoglobin?

- (1) High H^+ concentration and higher temperature
- (2) Low pO_2 and high pCO_2
- (3) High pO_2 and higher temperature
- (4) High pH and lower temperature

141. How many of the substances given in the box below are reabsorbed actively by the tubular epithelial cells in the PCT of nephron?

Glucose, amino acids, Na^+ , urea, water

Select the correct option.

- (1) Three
- (2) Two
- (3) Four
- (4) Five

142. Which of the following does not participate in the formation of filtration membrane through which ultrafiltration occurs in a nephron?

- (1) The epithelium of Bowman's capsule
- (2) Basement substance
- (3) Brush border epithelium of collecting duct
- (4) The endothelium of glomerular blood capillaries

143. Select the correct option to complete the analogy w.r.t. humans

Transport of O_2 by RBCs in blood : 97% :: Transport of CO_2 by RBCs in blood : ____

- (1) 70%
- (2) 3%
- (3) 20-25%
- (4) 7%

144. Kidneys are situated between the levels of ____ vertebrae close to the dorsal inner wall of the abdominal cavity. Select the correct option to fill in the blank.

- (1) T_{10} and L_1
- (2) T_{12} and L_3
- (3) L_5 and S_1
- (4) T_{10} and L_2

145. Assertion (A): The human heart is myogenic.

Reason (R): The nodal musculature of human heart has the ability to generate action potentials without any external stimuli.

In the light of above statements, select the correct option.

- (1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (2) (A) is true but (R) is false
- (3) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (4) Both (A) and (R) are false

146. Read the following statements carefully.

Statement A: Lungs possess alveoli which do not collapse even after forceful expiration.

Statement B: The volume of air remaining in the lungs even after a forcible expiration is called residual volume.

- (1) Both statements A and B are correct
- (2) Only statement A is correct
- (3) Both statements A and B are incorrect
- (4) Only statement B is correct

147. In cortical nephrons,

- (1) DCT is absent whereas PCT is present
- (2) Efferent arteriole emerging from glomerulus forms loop of Henle
- (3) Loop of Henle is very long and runs deep into the medulla
- (4) Vasa recta is absent or highly reduced

148. Which of the following options represents the location of regulatory centre that reduces the duration of inspiration and moderate the functions of respiratory rhythm centre?

- (1) Medulla oblongata
- (2) Pons
- (3) Cerebellum
- (4) Cerebrum

149. In humans, cartilage is absent in

- (1) Initial bronchioles
- (2) Sound box
- (3) Epiglottis
- (4) Alveoli

150. Select the **correct** match.

(1)	Branchial respiration	–	Adult <i>Rana tigrina</i>
(2)	Cutaneous respiration	–	<i>Pheretima</i>
(3)	Pulmonary respiration	–	Prawn
(4)	Cutaneous respiration	–	Human

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

151. Match column I and column II w.r.t. functions of renal tubules.

Column I		Column II	
(a)	Descending limb of loop of Henle	(i)	Maximum reabsorption of water and electrolytes
(b)	Distal convoluted tubule	(ii)	Conditional reabsorption of Na^+ and H_2O
(c)	Ascending limb of loop of Henle	(iii)	Concentrates the filtrate (from 300 mOsmolL^{-1} to $1200 \text{ mOsmolL}^{-1}$)
(d)	Proximal convoluted tubule	(iv)	Dilutes the filtrate (from $1200 \text{ mOsmolL}^{-1}$ to 300 mOsmolL^{-1})

Select the **correct** option.

- (1) a(i), b(ii), c(iii), d(iv)
- (2) a(iii), b(ii), c(iv), d(i)
- (3) a(iv), b(iii), c(ii), d(i)
- (4) a(iii), b(i), c(iv), d(ii)

152. Choose the **correct** option w.r.t cells that make 20-25% of the total leucocytes in a normal adult human

- (1) Play an important role in immune response of the body
- (2) Categorised as granulocytes
- (3) Phagocytose the foreign organisms
- (4) Secrete histamine and serotonin

153. Choose the **correct** statement.

- (1) In mammals, right atrium receives the oxygenated blood from pulmonary vein.
- (2) In fishes, gills perform the oxygenation of blood and deoxygenated blood enters the heart.
- (3) In birds, a single ventricle is present which pumps out mixed blood.
- (4) In amphibians, the left atrium receives deoxygenated blood.

154. In hepatic portal system, the hepatic portal vein carries blood from

- (1) Kidney to the liver
- (2) Intestine to the liver
- (3) Liver to heart
- (4) Liver to intestine

155. In humans, systemic circulation **does not** include

- (1) Left ventricle
- (2) Aorta
- (3) Left atrium
- (4) Right atrium

156. Stroke volume can be best described as

- (1) Volume of blood that remains in each ventricle after ventricular systole.
- (2) Volume of blood pumped out by each ventricle of heart per minute
- (3) Cardiac output $\times$ Heart rate
- (4) Volume of blood pumped out by each ventricle during a cardiac cycle

157. Which among the following is a **correct** match?

(1)	Heart attack	–	When enough blood is pumped out by the heart to meet the needs of the body
(2)	Angina	–	Congestion of lungs
(3)	Cardiac arrest	–	Heart stops beating
(4)	Atherosclerosis	–	Decreases thickness of blood vessels

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

158. Which of the following conditions/factors can decrease the cardiac output?

- (1) Neural signals through parasympathetic nerves
- (2) Strenuous exercise and workout
- (3) Neural signals through sympathetic nerves
- (4) Release of adrenal medullary hormones

159. In a standard ECG, which of the following events occurs just after the Q wave?

- (1) Depolarisation of the atria
- (2) Repolarisation of the ventricles
- (3) Initiation of ventricular contraction
- (4) End of ventricular systole

160. Which of the following is **not** a part of renal tubule?

- (1) A hairpin shaped Henle's Loop
- (2) Bowman's capsule
- (3) A straight tube in which DCTs open
- (4) PCT and DCT

161. For hemodialysis of uremic patients, all of the following are present in the dialysing fluid, **except**

- (1) Nitrogenous wastes
- (2) Glucose
- (3) Electrolytes
- (4) Water

162. Match column I with column II w.r.t. animals and their excretory structures.

Column I		Column II	
(a)	<i>Amphioxus</i>	(i)	Nephridia
(b)	<i>Pheretima</i>	(ii)	Protonephridia
(c)	Prawn	(iii)	Malpighian tubules
(d)	<i>Periplaneta</i>	(iv)	Green glands

Select the **correct** option.

- (1) a(i), b(iii), c(ii), d(iv)
- (2) a(ii), b(i), c(iv), d(iii)
- (3) a(i), b(iv), c(iii), d(ii)
- (4) a(iv), b(iii), c(i), d(ii)

163. Select the **correct** statement among the following w.r.t adult human under normal physiological conditions.

- (1) On an average, 35 – 60 gm of urea is excreted out per day
- (2) In 100 mL of blood, 12 – 16 gm of haemoglobin is present
- (3) 2.0 to 2.5 litres of urine is excreted out per day
- (4) Average breathing rate is 22 – 36 times min^{-1}

164. In hemodialysis, blood drained from a convenient artery is pumped into a dialysing unit after adding 'heparin' in it. Select the **correct** option which is true for heparin.

- (1) Released by only cells that are devoid of nucleus
- (2) Secreted by granulocytes that make up 0.5 to 1% of the total WBCs
- (3) It is a coagulant, present in the serum of blood
- (4) Formed by eosinophils in red bone marrow of humans

165. The process of release of urine involves

- (1) Contraction of the urethral sphincters
- (2) Activation of the stretch receptors present on the walls of the urinary bladder
- (3) Passing of sensory messages from CNS to urethra
- (4) Relaxation of smooth muscles of the urinary bladder

166. Afferent arteriole which forms glomerulus is a branch of

- (1) Peritubular capillaries
- (2) Renal artery
- (3) Renal vein
- (4) Vasa recta

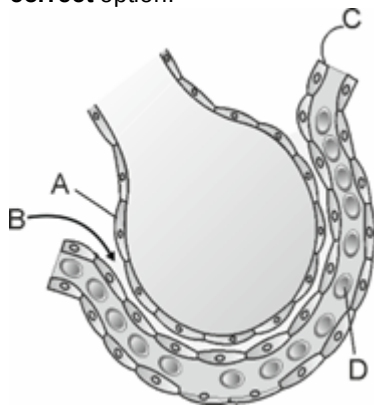
167. Maximally, how many molecules of O_2 are carried by one molecule of red coloured iron containing respiratory pigment present in human blood?

- (1) Six
- (2) Five
- (3) Four
- (4) Three

168. Under normal physiological conditions, in humans, every 100 mL of oxygenated blood can deliver around **X** mL of O_2 to the tissues whereas every 100 mL of deoxygenated blood delivers approximately **Y** mL of CO_2 to the alveoli. Identify **X** and **Y** respectively and select the **correct** option.

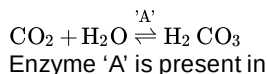
- (1) 4, 5
- (2) 5, 4
- (3) 10, 20
- (4) 20, 10

169. Identify A, B, C and D w.r.t. below given figure and select the correct option.



- (1) A – Endothelium of blood capillary
- (2) B – Squamous epithelium of alveolar wall
- (3) C – Basement substance
- (4) D – RBCs in blood capillary

170. Observe the following reaction.



- (1) WBCs and blood plasma
- (2) RBCs and blood plasma
- (3) Platelets and serum
- (4) RBCs and WBCs

171. **Assertion (A):** Terrestrial adaptation necessitated the production of less toxic nitrogenous wastes for conservation of water.

Reason (R): Elimination of most toxic nitrogenous waste requires least amount of water for its elimination.
In light of the above statements, select the correct option.

- (1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (2) Both (A) and (R) are false
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are true and (R) is the correct explanation of (A)

172. In human kidney, the 'X' are arranged in an intricate manner which leave some minute spaces called filtration slits or slit pores. Here 'X' represents

- (1) Endothelial cells of glomerulus
- (2) Squamous epithelial cells of Henle's loop
- (3) Cuboidal cells of DCT
- (4) Epithelial cells of Bowman's capsule called podocytes

173. The condition 'erythroblastosis foetalis' can be avoided by administering 'X' to the Rh –ve mother immediately after 'Y'. Select the option which correctly represents 'X' and 'Y' respectively.

- (1) Rh antigen, Conception for the 2nd time
- (2) Rh antibodies, Delivery of the 1st Rh –ve child
- (3) Anti-Rh antigens, pregnancy for the first time
- (4) Anti-Rh antibodies, Delivery of the 1st Rh+ve child

174. Select the correct statement w.r.t lymph.

- (1) It carries all the proteins present in the blood.
- (2) It has same mineral distribution as that of blood plasma.
- (3) It is devoid of lymphocytes and immune cells.
- (4) It is collected by lymphatic system that drains it back into the major arteries.

175. Chordae tendinae in the human heart prevents

- (1) Flow of blood from left atrium to right atrium
- (2) Back flow of blood from right atrium to vena cava
- (3) Back flow of blood from ventricles to atria
- (4) Flow of blood from left ventricle to aorta

176. All of the following are basis of differences between veins and arteries, except

- (1) Presence of endothelium
- (2) Direction of blood flow
- (3) Diameter of lumen
- (4) Presence of valves

177. Read the following statements.

Statement A: Both ADH and angiotensin II are vasoconstrictors and have same effects on blood vessels which cause increase in the glomerular blood pressure.

Statement B: Aldosterone is released from adrenal cortex whereas ADH is secreted from hypothalamus.
Select the correct option.

- (1) Both statements A and B are correct
- (2) Both statements A and B are incorrect
- (3) Only statement A is correct
- (4) Only statement B is correct

178.How many of the following statements is/are **correct**?

- (a) Exchange part of human respiratory system is the site of diffusion of O_2 and CO_2 between blood and atmospheric air.
- (b) Pleural fluid present in between the double layered pleura reduces friction on the lung surface.
- (c) Alveoli and their ducts are not the part of conducting zone of the respiratory system.
- (d) In humans, any change in the volume of thoracic cavity will be reflected in pulmonary cavity.

Select the **correct** option.

- (1) One
- (2) Two
- (3) Three
- (4) Four

179.Match column I with column II.

Column I		Column II	
(a)	IC	(i)	ERV + RV
(b)	FRC	(ii)	TV + IRV
(c)	TLC	(iii)	VC + RV
(d)	VC	(iv)	ERV + TV + IRV

Select the **correct** option.

- (1) a(iv), b(ii), c(iii), d(i)
- (2) a(ii), b(i), c(iii), d(iv)
- (3) a(i), b(ii), c(iii), d(iv)
- (4) a(iv), b(iii), c(i), d(ii)

180.Which of the following factors primarily cannot affect the rate of diffusion of gases at alveoli?

- (1) The reactivity of gases
- (2) Solubility of the gases
- (3) Thickness of the membranes involved in diffusion
- (4) Concentration gradient of gases



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